Thermodynamics and Statistical Physics

4th Exercise Class

Heat Engines and 2nd Law; Thermodynamic Temperature Scale; Carnot Cycle; Otto Cycle

Exercises for the Class

1. A particular heat engine has a mechanical power output of 5 kW and an efficiency of 25%. The engine expels $9 \cdot 10^3$ J of exhaust energy in each cycle. Find

   (a) the energy taken in during each cycle;

   (b) the time interval for each cycle.

2. A certain amount of ideal gas performs the $ABCDA$ cycle shown in Fig. 1.

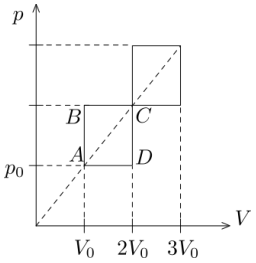


Fig. 1

   (a) How does the thermal efficiency of the cycle change if the rectangle representing the cycle is shifted as shown in the figure?

   (b) What is the ratio of the two efficiencies?

3. An inventor claims to have developed an engine that takes in $1.1 \cdot 10^8$ J at 400 K, rejects $5.0 \cdot 10^7$ J at 200 K, and delivers 16.6 kW hours of work. Would you advise investing money in this project?

4. Which gives the greater increase in the efficiency of a Carnot engine: increasing the temperature of the hot reservoir or lowering the temperature of the cold reservoir by the same amount?

5. An idealized diesel engine operates in a cycle known as the air-standard diesel cycle shown in Fig. 2. Fuel is sprayed into the cylinder at the point of maximum compression, B. Combustion occurs during the expansion $B \to C$, which is modeled as an isobaric process. Show that the efficiency of an engine operating in this idealized diesel cycle is

$$e = 1 - \frac{1}{\gamma}\frac{T_D - T_A}{T_C - T_B}.$$

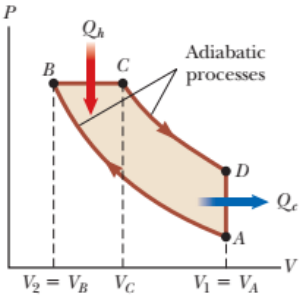


Fig. 2

6. Suppose you build a two-engine device with the exhaust energy output from one heat engine supplying the input energy for a second heat engine. We say that the two engines are running in series. Let e_1 and e_2 represent the efficiencies of the two engines.

   (a) The overall efficiency of the two-engine device is defined as the total work output divided by the energy put into the first engine by heat. Show that the overall efficiency e is given by

$$e = e_1 + e_2 - e_1 e_2$$

   (b) Assume that the two heat engines are Carnot engines. The first engine operates between temperatures T_1 and T_2. In the second, the gas temperature varies between T_2 and T_3. Express the efficiency of the whole system in terms of the temperatures.

7. Two Carnot engines have the same efficiency. One engine works as a heat pump in the reverse direction, the the other as a refrigerator in the reverse direction. The heat pump has a coefficient of performance 1.5 times that of the refrigerator. Determine:

   (a) the coefficient of performance of the refrigerator;

   (b) the coefficient of performance of the heat pump;

   (c) the efficiency of the heat engines.

Homework to Be Submitted

1. We want to cool down 1 kg of water from its original temperature 30 °C to 10 °C with an ideal refrigerator (Carnot-cycle). During the cooling process the temperature of the environment remains constant 30 °C. What is the amount of work needed for this process? The heat conduction between the water and the environment is negligible, the specific heat capacity of water is 4200 J/(kg °C).

Homework for Practice

1. The p-V diagram in Fig. 3 shows two cycles. The cycles are taken by monatomic ideal gases ($\gamma = = 5/3$), and both are clockwise. The lower cycle has an efficiency of 12.5%.

   (a) What is the value of x?

   (b) What is the efficiency of the upper cycle?

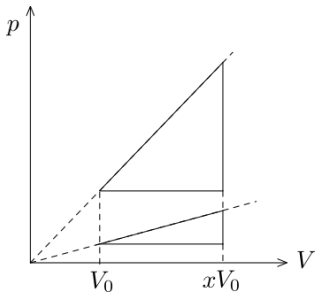


Fig. 3

2. A heat pump used for heating shown in Fig. 4 is essentially an air conditioner installed backward. It extracts energy from colder air outside and deposits it in a warmer room. Suppose the ratio of the actual energy entering the room to the work done by the device's motor is 10% of the theoretical maximum ratio. Determine the energy entering the room per joule of work done by the motor given that the inside temperature is 20 °C and the outside temperature is −5 °C.

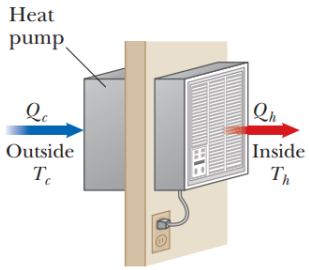


Fig. 4

3. An electric generating station is designed to have an electric output power of 1.40 MW using a turbine with two-thirds the efficiency of a Carnot engine. The exhaust energy is transferred by heat into a cooling tower at 110 °C.

   (a) Find the rate at which the station exhausts energy by heat as a function of the fuel combustion temperature T_h.

   (b) If the firebox is modified to run hotter by using more advanced combustion technology, how does the amount of energy exhaust change?

   (c) Find the exhaust power for $T_h = 800$ °C.

   (d) Find the value of T_h for which the exhaust power would be only half as large as in part (c).

   (e) Find the value of Th for which the exhaust power would be one-fourth as large as in part (c).

4. For an ideal gas, we perform the following cycle. Initially, we increase the pressure of the gas with temperature T_1 at constant volume until its temperature reaches $T_2 > T_1$. Then we adiabatically expand the gas until its temperature returns to T_1, and finally, by isothermally compressing it, we return the gas to its initial state.

   (a) Plot the process on a p–V diagram.

   (b) Determine the efficiency of the cycle as a function of T_1 and T_2.